FOSS × BIS HACKATHON

AI-Powered Website Chatbot Challenge

Problem Statement, Test Cases & Judging Criteria — 24-Hour Sprint

1. Challenge Overview

You have 24 hours to build an AI-powered chatbot that can answer any question about the BIS website (bis.gov.in) — accurately, conversationally, and with source citations. The core technology is a RAG (Retrieval-Augmented Generation) pipeline that you will build from scratch.

The chatbot must crawl, index, and understand content from across the full BIS website — including standards, certification schemes, laboratory services, consumer programmes, publications, press releases, and more — and surface precise answers from anywhere on the site.

The BIS Website — Your Knowledge Source

The Bureau of Indian Standards (bis.gov.in) is India's national standards body. Its website spans dozens of pages covering product certification schemes, laboratory accreditation, standards development, consumer affairs, international relations, publications, legislation, and more. Your chatbot must index and understand all of it.

2. Problem Statement

The BIS website currently has no intelligent assistant. Users searching for certification processes, scheme details, fee structures, or policy information must manually navigate through multiple pages, skim through content, and piece together answers themselves. This creates three distinct problems:

🔍 **Problem 1 — Search Returns Links, Not Answers**

Traditional keyword search returns a list of pages. Users must read, skim, and guess which section contains what they need.

⏱ **Problem 2 — Content is Scattered Across Many Pages**

The BIS website spans dozens of pages across standards, schemes, labs, consumer programmes, publications, and more. Answers are often split across multiple sections.

😤 **Problem 3 — Frustration Drives Disengagement**

When users cannot find answers, they raise support queries or abandon the site entirely. Poor discoverability is a silent driver of disengagement.

3. What You Are Building

A drop-in chatbot widget powered by a custom RAG pipeline that ingests and indexes the full BIS website and answers user questions with grounded, cited responses.

Required Pipeline

Stage	Name	What Happens
1	**Ingest**	Recursively crawl all pages across bis.gov.in — standards, certification, labs, publications, news, FAQs, consumer programmes
2	**Chunk**	Split content into ~500-token overlapping passages; preserve headings and structural units
3	**Embed**	Convert chunks to vector representations using an embedding model (e.g. text-embedding-3-small)
4	**Store**	Index vectors + metadata (URL, title, content type, timestamp) in a vector database
5	**Retrieve**	Find the top-K most relevant chunks for any user query via cosine similarity search
6	**Answer**	Pass retrieved context to an LLM with a grounding prompt; stream the response to the UI

Required Deliverables

- GitHub repository with all code written during the hackathon
- README.md with setup instructions, architecture overview, and tools used
- Live deployed demo (Vercel, Railway, Render, or equivalent)
- Architecture diagram showing the full RAG pipeline
- Working chatbot indexed on real BIS website content

4. Open Test Cases — Evaluated Live During Demo

These 5 test cases are known to you in advance. Your chatbot will be asked these exact questions (or close variants) during the live demo. Each is worth 4 points. Judges will verify answer accuracy, source citation quality, and response behaviour.

How Judging Works

During your demo, judges will type each query directly into your chatbot. They will assess: (a) whether the answer is factually correct and grounded in BIS website content, (b) whether a valid source URL is cited, and (c) whether the chatbot behaves gracefully under edge conditions.

#	Scenario	What the Chatbot Must Do	Pass Condition
TC-01	Basic Fact RetrievalQuery: "What is BIS and what are its core functions?"	Retrieve relevant content from the About / Overview section of the BIS website and return an accurate, concise answer grounded in that content.	Answer is factually correct, cites source URL, contains no fabricated claims.

#	Scenario	What the Chatbot Must Do	Pass Condition
TC-02	Standards & CertificationQuery: "How do I apply for a BIS certification for my product?"	Locate the certification / scheme pages, extract the correct process steps, and present them in a readable, structured format with source links.	Correct process outlined, at least one source citation, no hallucinated requirements.
TC-03	Multi-turn ConversationTurn 1: "What schemes does BIS offer?"Turn 2: "Tell me more about the third one."	Maintain conversation context across turns. Turn 2 must resolve 'the third one' without the user restating the topic.	Turn 2 correctly references the scheme named in Turn 1. No context loss.
TC-04	Cross-Section RetrievalQuery: "What is BIS doing in the area of consumer awareness?"	Pull relevant content from multiple sections (e.g. press releases, programmes, publications) and synthesise a coherent answer.	Answer draws from 2+ content types, cites distinct source URLs, answer is coherent.
TC-05	Graceful Out-of-Scope HandlingQuery: "What is the current stock price of Tata Steel?"	Recognise the query is outside the scope of the indexed BIS website and respond gracefully without fabricating an answer.	Chatbot declines to answer, explains it can only answer from BIS website content, suggests a relevant BIS topic if possible.

5. Hidden Test Cases — Revealed After Submission

These 10 additional scenarios will be evaluated after submissions close. You will not see the exact queries in advance. They are designed to stress-test the depth of your RAG pipeline across harder retrieval scenarios, multi-turn memory, and adversarial probes.

Why Hidden Test Cases?

The hidden cases reward teams who build genuinely robust systems rather than demo-polished prototypes. They test cross-page synthesis, hallucination resistance, precise fact retrieval, multi-turn memory, and handling of ambiguous or adversarial queries. Each hidden case is worth 2 points (20 points total).

#	Domain	Question Type	Why It's Hard
HTC-01	Laboratory Services	Specific service lookup	Requires navigating nested lab/testing pages
HTC-02	Legislation & Acts	Policy/legal content	Legal language needs precise retrieval, no paraphrase errors
HTC-03	International Relations	Cross-page synthesis	Content spread across multiple international cooperation pages
HTC-04	Grievance / Complaints	Process + contact info	Tests whether contact details and steps are retrieved together
HTC-05	Publications & Reports	Document discovery	Must surface specific reports, not just the publications index
HTC-06	Hallucination probe	Invented regulation check	Query references a non-existent BIS rule — bot must not fabricate
HTC-07	Multilingual intent	English query, Hindi-labelled content	Tests robustness when indexed content has mixed-language labels

#	Domain	Question Type	Why It's Hard
HTC-08	Numerical/date facts	Fee or timeline lookup	Precise numeric facts must match source exactly
HTC-09	Follow-up pronoun resolution	Multi-turn (3 turns)	Third turn uses ambiguous pronoun — context must be retained
HTC-10	Comparative query	Side-by-side scheme comparison	Requires retrieving and comparing content from two separate pages

Note: Exact queries for each hidden test case will be released to all teams simultaneously immediately after the submission deadline. Results will be evaluated by the judging panel.

6. Scoring Breakdown

Category	Points	Notes
Technical Implementation	25	End-to-end RAG pipeline, ingestion quality, retrieval accuracy
Answer Quality & Accuracy	25	Grounded answers, zero hallucinations, correct citations
5 Open Test Cases	20	4 points each — judged live during demo
10 Hidden Test Cases	20	2 points each — evaluated post-submission
Innovation & UX	10	Creative features, UI polish, unique approach
TOTAL	**100**	

Tip: The open and hidden test cases together account for 40 points — nearly half the total score. Teams who invest in retrieval quality, grounding, and citation accuracy will have a significant advantage.

7. What Judges Look For

7.1 Technical Implementation (25 pts)

- Does the ingestion pipeline handle the full breadth of content types on the BIS website?
- Does semantic retrieval surface genuinely relevant chunks — not just keyword matches?
- Is the LLM answer strictly grounded in retrieved context, with no fabrication?
- Is the pipeline architecturally sound and maintainable?

7.2 Answer Quality & Accuracy (25 pts)

- Every claim in the answer must be traceable to a retrieved chunk from the BIS website.
- Zero hallucinations — if the answer is not in the indexed content, the bot must say so.
- Source citations must link to the actual page where the content was found.

- Multi-turn context must be maintained coherently across the conversation.

7.3 Test Case Performance (40 pts)

- 5 Open Cases (4 pts each): Evaluated live during the demo presentation.
- 10 Hidden Cases (2 pts each): Evaluated by the panel after submission closes.
- Partial credit may be awarded for partially correct or well-reasoned answers.

7.4 Innovation & UX (10 pts)

- Novel retrieval strategies: hybrid search, re-ranking, query rewriting.
- UI polish: streaming responses, readable citations, content-type labels.
- Creative features: follow-up suggestions, scope filters, feedback mechanism.

8. 24-Hour Timeline

Time Window	Phase	Goal
Hour 0–3	Setup & Scaffolding	Project init, API keys, vector DB index, site crawl plan
Hour 3–10	Ingestion Pipeline	Crawler, chunking, embedding, vector upsert — full BIS site indexed
Hour 10–16	Query API	Semantic search endpoint, prompt engineering, streaming response
Hour 16–20	Chat Widget	Frontend component, streaming display, source citations
Hour 20–22	Polish & Testing	Edge cases, cross-section retrieval quality, mobile responsiveness
Hour 22–23	Demo Prep	Final deploy, slides, rehearsal against the 5 open test cases
Hour 24	Submission + Presentations	GitHub repo submitted; live demo to judges

The 5 open test cases are published at Hour 0. Use them to guide your development and prioritise retrieval quality from the start.

9. Rules Summary

- Teams of 2–5 registered participants only.
- All code must be written during the hackathon window (Hour 0 to Hour 24).
- The chatbot must be indexed on real BIS website content (bis.gov.in).
- A working live demo must be presented — pre-recorded video alone is not accepted.
- Any public LLM provider, vector database, and open-source library is permitted.
- Starter code or boilerplate created before the event is not permitted.
- Source code must be submitted via the designated GitHub repository before the deadline.

10. Suggested Tech Stack(Optional)

The following is a recommendation, not a requirement. Use whatever tools let your team build the best system.

Layer	Recommended Options
LLM	Gemini 2.5,Llama 3,Mistral 7B,Mixtral 8x7B,Phi-3,Gemma
Embeddings	text-embedding-3-small, all-MiniLM-L6-v2
Vector Database	Pinecone, Weaviate, Qdrant, pgvector
Site Crawler	Crawlee, Scrapy, Firecrawl, sitemap parsers
RAG Framework	LangChain.js, LlamaIndex, Haystack
Backend Framework	Next.js 15, Remix, FastAPI
Streaming / Memory	Redis, Upstash, Server-Sent Events (SSE)
UI Components	Tailwind CSS, shadcn/ui
Deployment	Vercel, Railway, Render

Good luck. Build something great.

FOSS × BIS Hackathon — AI Website Chatbot Challenge — 24 Hours

Draft Standards Preparation Activity

Pre-Standardisation Study

Topic : Performance and Accuracy of Fitness Wearables

Submission link : https://forms.gle/Us5zR6FFGeNevcSs6

Submission deadline : 12 March 2026

Activity Overview

Participants will take part in a **Draft Standards Preparation Activity**, simulating the **standard development process followed by the Bureau of Indian Standards (BIS)**.

Each team will prepare a **Pre-Standardisation Study Report** proposing draft standards for the given topic.

The objective of this activity is to help participants understand:

- How **product quality standards are developed**
- How **performance benchmarks are defined**
- How **testing methodologies are proposed**

Topic for the Study

Performance and Accuracy of Fitness Wearables

Participants will study consumer wearable devices. These include:

- **Smartwatches** *(e.g. Apple Watch, Samsung Galaxy Watch):* multifunctional devices that track health metrics alongside notifications and apps
- **Fitness Bands** *(e.g. Mi Band, Fitbit):* simpler, dedicated devices focused on activity and health tracking
- **Health Monitoring Wearables** *(e.g. pulse oximeters, ECG patches):* devices designed specifically to monitor medical or clinical health parameters

The goal is to propose standard guidelines for measuring how well these devices actually perform.

In other words: when your watch says your heart rate is 72 BPM or you've walked 8,000 steps, how do we know if that's true?

Your report should define:

- **What** should be tested (e.g. heart rate accuracy, step counting)
- **How** it should be tested (e.g. comparing against medical-grade equipment)
- **What counts as acceptable** (e.g. heart rate readings within ±5 BPM of the actual value)

Think of it as writing the rulebook that manufacturers would have to follow before claiming their device is accurate.

What Participants Need To Do

Participants must prepare a **technical report** that includes:

1. Understanding the product category.
2. Identifying **key performance parameters**.
3. Studying existing practices or issues in the market.
4. Proposing **testing methods**.
5. Suggesting **performance benchmarks for standardization**.

Suggested Report Structure

Participants are advised to follow the structure below.

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1. Title Page

Example:

Pre-Standardisation Study on Performance and Accuracy of Fitness Wearables

Prepared by:
Team Name / Participants

Event:
Draft Standards Preparation Activity

Organized by:
FOSS NIT Srinagar in collaboration with **BIS**

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2. Introduction

Explain:

- What **fitness wearables** are
- Their **role** in health and fitness monitoring
- Why **standardization is important**

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3. Scope of the Study

Define which devices are included.

Example:

- Smartwatches

- Fitness bands
- Health monitoring wearables

Parameters considered:

- Heart rate monitoring
- Step counting
- Sleep tracking
- GPS tracking
- Battery performance

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4. Existing Practices

Participants may briefly discuss:

- Current market practices
- Accuracy claims by manufacturers
- Problems in measurement reliability

5.Key Performance Parameters

Examples include:

- Heart Rate Accuracy
- Step Counting Accuracy
- Sleep Tracking Reliability
- GPS Accuracy
- Battery Life

6.Proposed Testing Methodology

Participants should describe **how each parameter should be tested**.

Example:

Heart Rate Testing
Comparison with **medical grade ECG devices**

Step Count Testing
Compare **actual steps vs device recorded steps**

Sleep Tracking
Compare wearable data with **sleep monitoring systems**

7.Proposed Performance Benchmarks

Participants should define acceptable limits.

Example:

Heart Rate Accuracy
±5 BPM

Step Count Accuracy
±3% error margin

Battery Life
Minimum 24 hours operation

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8. Data Reliability and Safety

Discuss:

- Sensor calibration
- Data reliability
- Privacy considerations

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9. Conclusion

Summarize:

- Importance of standardization
- Benefits for **consumers and manufacturers**

10.Submission Instructions :

Participants must submit their **report in PDF format**.

Submission will be done through the **Google Form shared via email**.

Submission Link:
https://forms.gle/Us5zR6FFGeNevcSs6

The submission deadline is 12 March 2026.

Important Rules

- The report will be **checked for plagiarism and AI-generated content**.
- The similarity score **must be below 20%**.
- Submissions exceeding the limit may be **disqualified**.

Evaluation Criteria

Submissions will be evaluated based on:

- Understanding of the topic
- Quality of proposed standards

- Technical clarity
- Practical testing methods
- Structure and presentation

Prizes

1st Prize - ₹2,000

2nd Prize - ₹1,500

3rd Prize -₹1,000

Certificates will be provided to **all participants!**

Reference Material

Participants should refer to the **attached Pre-Standardisation Study case studies** shared in the drive link with this document. These references provide an idea of:

- Structure of a standard study
- Level of technical detail expected
- How standards are proposed

Drive Link:

https://drive.google.com/drive/folders/1GPTRElPtDFSpJA0gA4nU3i4jFcRg_7J8?usp=drive_link